ARCHITECTURE PORTFOLIO

BETTY KIBREAB

BA (Hons) ARCHITECTURE | 2023–2026

CV

About me

I am a BA (Hons) Architecture graduate from the University of Greenwich with a passion for creating meaningful, people-centred spaces. I am interested in exploring how design can respond to environmental challenges and transform them into opportunities for innovative and sustainable solutions.
This portfolio showcases a selection of my academic and personal work, reflecting my design process, technical skills, and creative approach, with a focus on clear communication and creating spaces that positively impact people and communities.

Contacts

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Education

BA (Hons) Architecture
University of Greenwich, London · **Sep 2023 – 2026**

Work Experience

Internship at Adjaye Associates, London · Jul 2019
Worked alongside architects and design team members, assisting with tasks and producing visual presentation materials.

Skills

Teamwork
Experience in group projects and collaborative design work.
Visualisations
Able to produce realistic visualisations that clearly communicate design concepts.
AI proficiency
Experience in applying AI tools to support design exploration, visualisation, and presentation.
Hand sketching
Ability to communicate and develop ideas through sketching.
Problem solving
Ability to develop design solutions in response to design constraints.
Research & Analysis
Experience in conducting site, precedent, and environmental research to guide design decisions.

Projects & Achievements

Residential Renovation Project · Aug 2024
Architecture summer course · Jul 2023
University of Brighton
Children's storybook Project · Nov 2017

Software

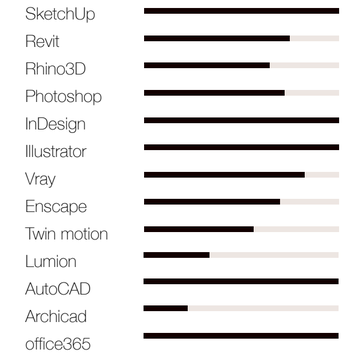


TABLE OF CONTENTS

01. **The Water Commons:**
Thames Bathhouse & Filtration Hub

02. **Sensus Restaurant**
Immersive Dining

03. **Potters Passage**
Pottery studio

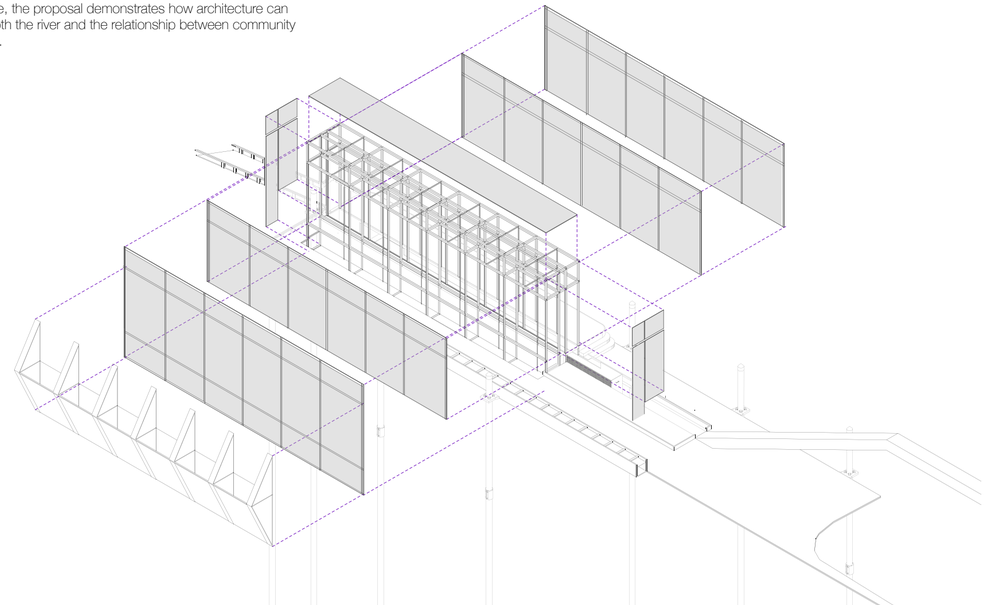
04. **McLaren Velocity Centre**
Formula 1 Experience Centre

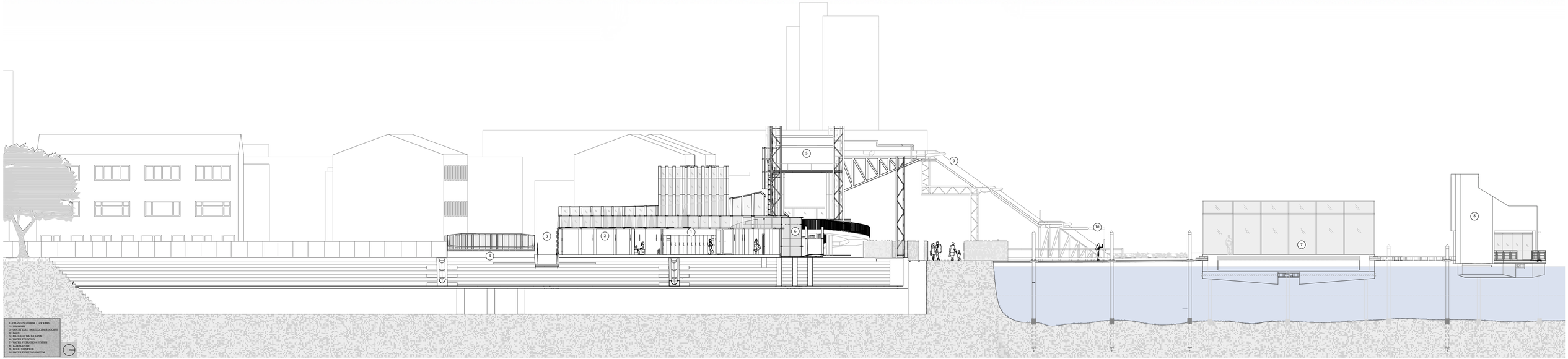
01 02 03 04

The Water Commons:

Thames Bathhouse & Filtration Hub

A floating civic hub that transforms river restoration into a shared public experience. Combining water filtration, scientific research, and a contemporary bathhouse, the project actively cleans the River Thames while reconnecting people with its ecological processes. By making environmental infrastructure visible and accessible, the proposal demonstrates how architecture can restore both the river and the relationship between community and water.

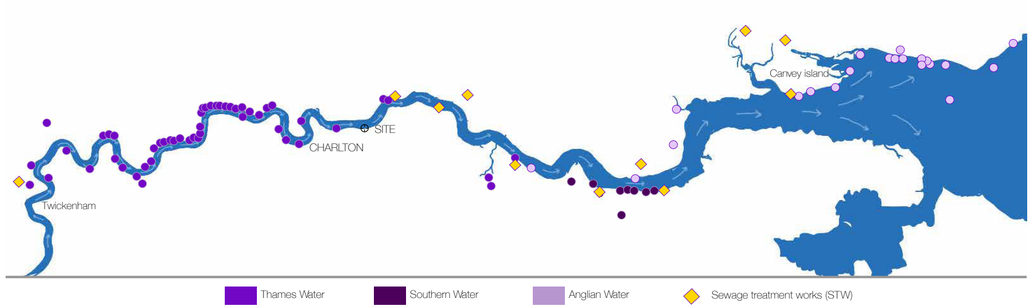


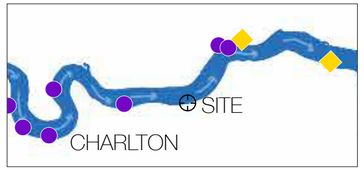

1 - CHANGING ROOM / LOCKERS
2 - COURTYARD
3 - COURTYARD / WHEELCHAIR ACCESS
4 - BATH
5 - RAINWATER WATER TANK
6 - WATER POLYCLEAN
7 - WATER FILTRATION SYSTEM
8 - LABORATORY
9 - REST CONVEYOR
10 - WATER PUMPING SYSTEM

SEWAGE OUTFLOWS INTO THE THAMES RIVER

DATA: Port of London Authority, 2025

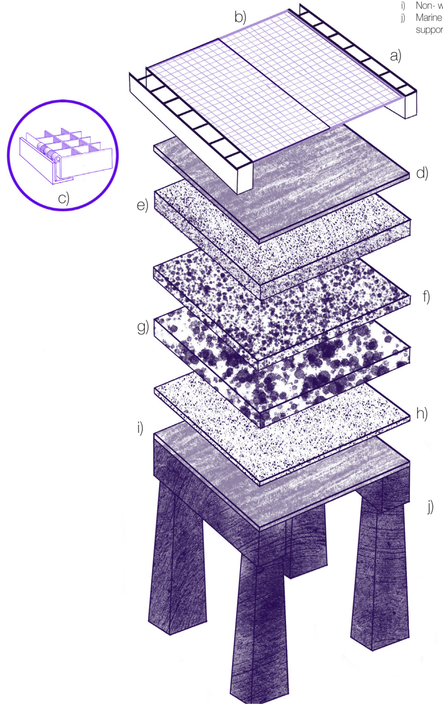
Closest outfall: Charlton



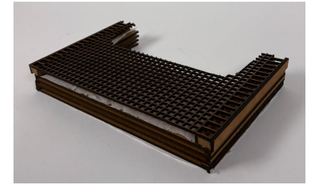


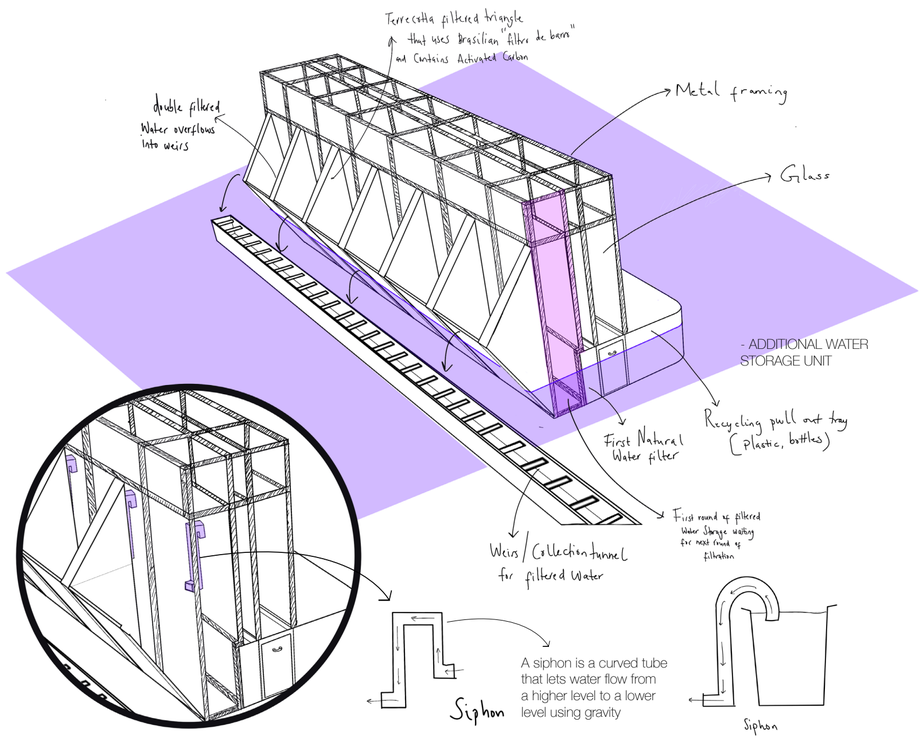
"The Charlton outfall releases untreated stormwater overflow during heavy rain. Tidal flow carries this pollution downstream toward Woolwich Dockyard."

a) Weirs/ surface collection tunnels -
b) GRP/FRP Grating -
c) Inspection hatches -
d) Non- woven Geotextile filter cloth (2-6mm)
e) FINE sand layer (100mm)
f) MEDIUM gravel layer (75mm)
g) COARSE gravel layer (150mm)
h) FINE gravel layer (20-30mm)
i) Non- woven Geotextile filter cloth (2-6mm)
j) Marine grade reinforced concrete dock/ support -

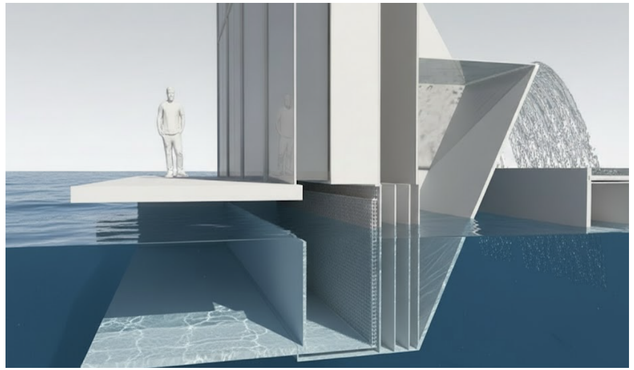


Filtration concept model

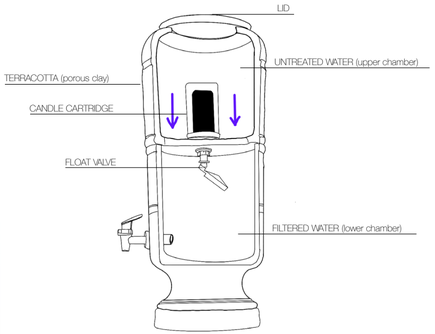




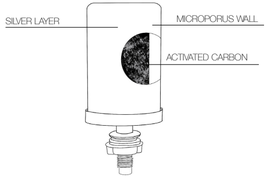
The filtration system works in three stages. First, water passes through a natural filter that removes sediment and larger impurities. It then flows through a terracotta filter inspired by the Brazilian filtro de barro, where activated carbon removes finer contaminants, odours, and chemicals. The purified water is transported through the water collection tunnels and stored in a clean water tank. During the process, plastic waste is separated, collected, and recycled, reducing pollution while producing clean water.



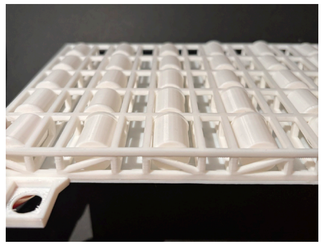
"FILTRO DE BARRO"

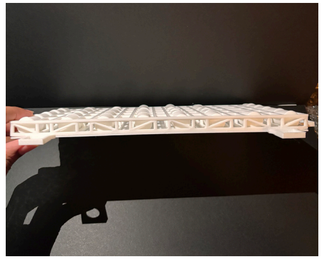


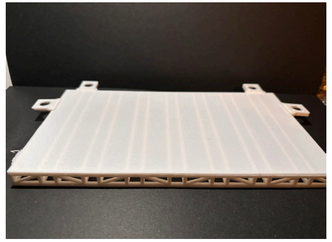
CANDLE CARTRIDGE

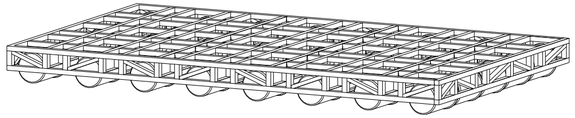


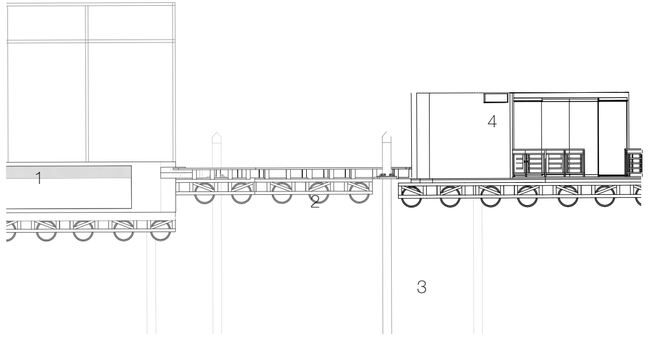
Outer Ceramic Shell
• Removes particles and blocks bacteria.
Silver Coating
• Prevents germs from growing.
Activated Carbon
• Improves taste and removes odors.







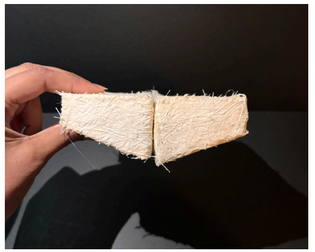


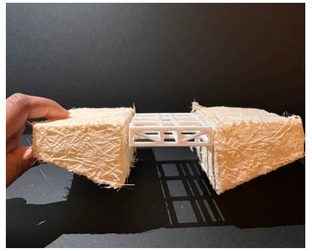


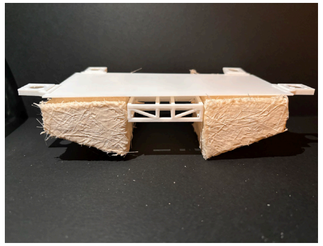
This test explores the use of air-filled barrels as a flotation system for the pontoon structure. Barrels were used to understand how trapped air provides buoyancy and how the system performs under load, investigating stability, load distribution, and movement in water.

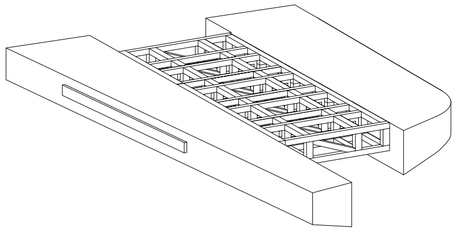
Findings:
- Barrels provide strong buoyancy due to trapped air
- Stability is limited when barrels are not evenly distributed
- Movement increases under uneven loads
- The system is bulky and less adaptable for controlled design
- Requires secure fixing to prevent shifting

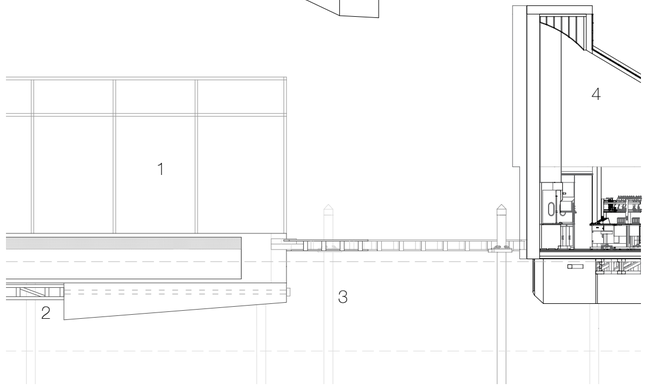
1 - filtration hub
2 - #1 floatation system
3 - guide piles
4 - water testing laboratory









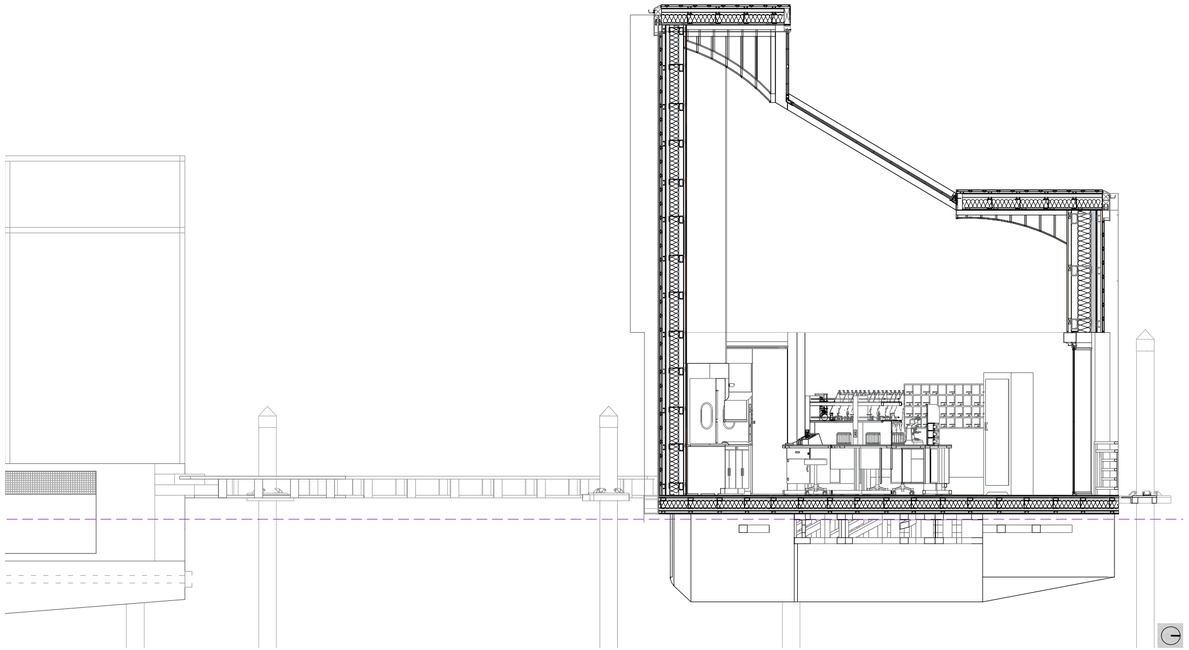


This test explores the flotation system The design was developed following the limitations of the barrel system, focusing on improving stability, control, and load distribution. using a resin coating, it created a harder and more fully sealed surface. This increased durability and resistance to water penetration, strengthening the overall flotation system.

Findings:
- resin coating created a stronger, fully sealed surface
- Improved waterproofing and structural durability
- Increased resistance to deformation in water conditions
- Catamaran system provides improved stability and balance
- Tester block remained level under applied load

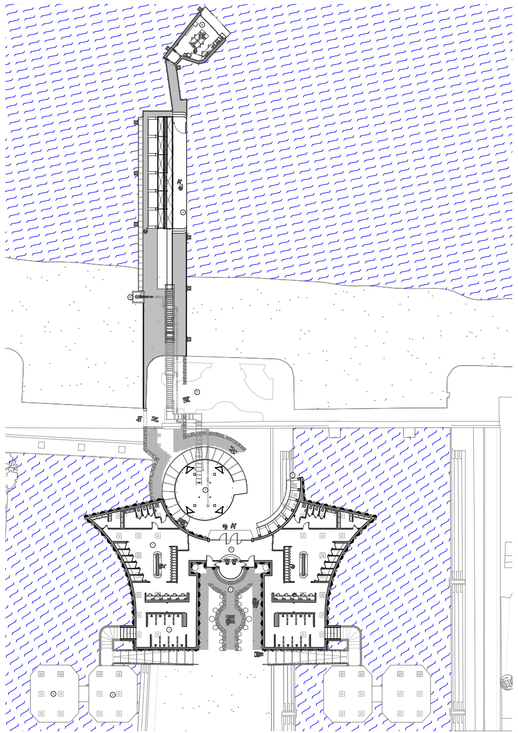
1- filtration hub
2- #2 floatation system
3- guide piles
4- water testing laboratory

1 - RECEPTION /ENTRANCE
2 - 2 STEP FILTERED WATER TANK
3 - RECYCLED PLASTIC STORAGE TANK
4 - WATER FOUNTAIN
5 - POOL
6 - WHEELCHAIR ACCESS RAMP

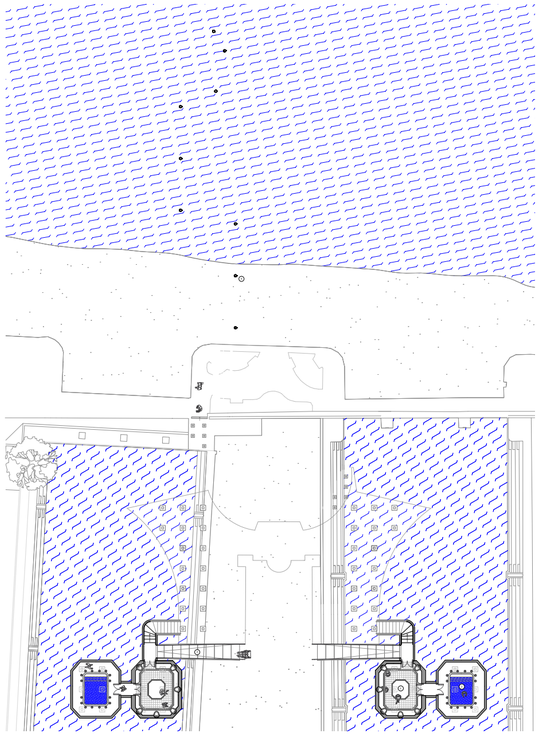


Technical section - Laboratory

1 - exterior wall design (marine-grade 316 grade stainless steel)	-	7 - primary insulation (rigid Polyisocyanurate)	150mm
2 - cladding backboard (plywood)	10mm	8 - SFS/ metsec structural frame (steel stud frame)	-
3 - ventilated air gap	40mm	9 - vapour control layer (polyethylene)	0.25mm
4 - sheating/ rigid panel (plywood panel)	12mm	10 - service void	35mm
5 - breather membrane	0.3mm	11 - internal lining (plasterboard)	15mm
6 - secondary insulation (thin rigid board)	60mm		



Ground floor



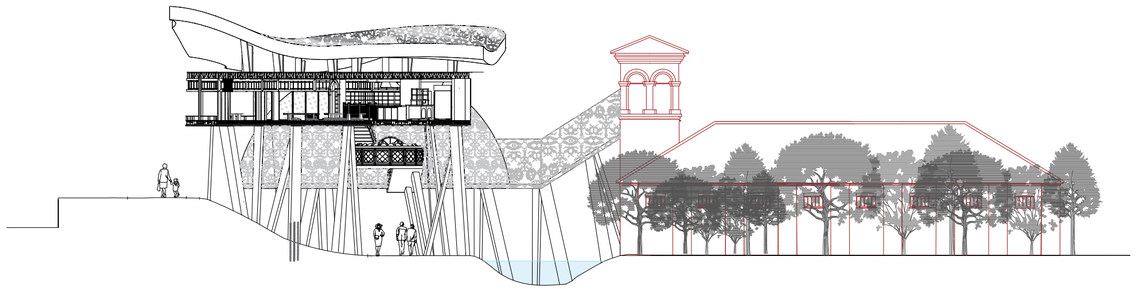
Basement floor

1 - changing rooms/ lockers
2 - reception
3 - showers
4 - courtyard/ wheel-chair access
5 - bath (hammam)
6 - pool
7 - filtered water tank
8 - water fountain
9 - water filtration system
10 - laboratory
11 - belt conveyor
12 - water pumping system

Sensus Restaurant

Immersive Dining

Sensus redefines humidity as an architectural resource, transforming Walthamstow's excess atmospheric moisture into renewable energy through a responsive façade. The harvested energy powers an interactive public installation and an immersive restaurant, where each space engages the six senses. Blending environmental technology with sensory design, the project demonstrates how architecture can convert climatic challenges into meaningful cultural and communal experiences.



Cross section

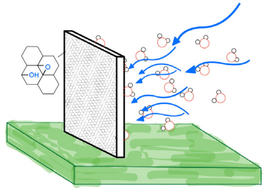
Here i tested a small-scale façade prototype made of a porous sponge layer, a graphene oxide slurry layer, and a sculptural façade with crevices inspired by William Morris.

The experiment investigated whether the layered system could absorb moisture from the air. A beaker of boiled water was used to simulate humidity. The prototype successfully absorbed and retained water vapour, confirming that the façade could capture atmospheric moisture and demonstrating the potential of graphene oxide as part of a passive, sustainable façade system.

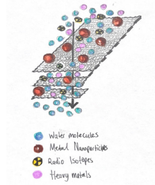
Graphene Oxide surface experiment

Graphene oxide (GO) is integrated into the façade system as a moisture-responsive material capable of absorbing water vapour directly from the atmosphere. Through the hydrovoltaic effect, the movement of water molecules across the GO surface generates a small electrical current, creating a potential renewable energy source without reliance on conventional fuels.

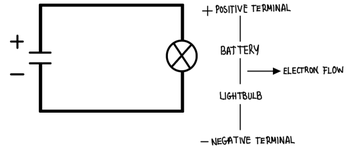
Water Absorption by Graphene

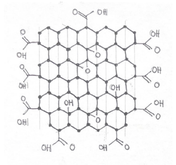


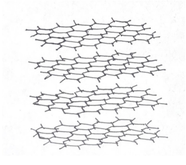
Interaction of Water Molecules with GO surfaces

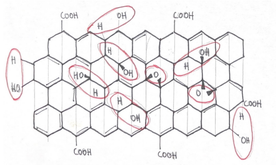


Electricity Generation (Hydrovoltaic Effect)

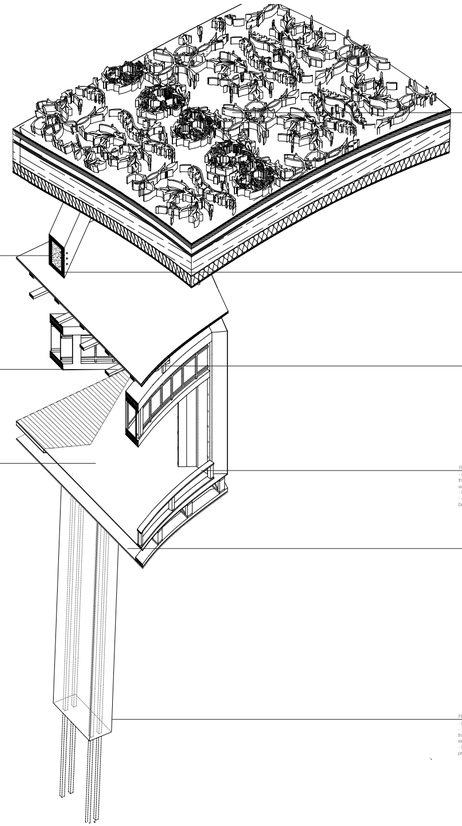


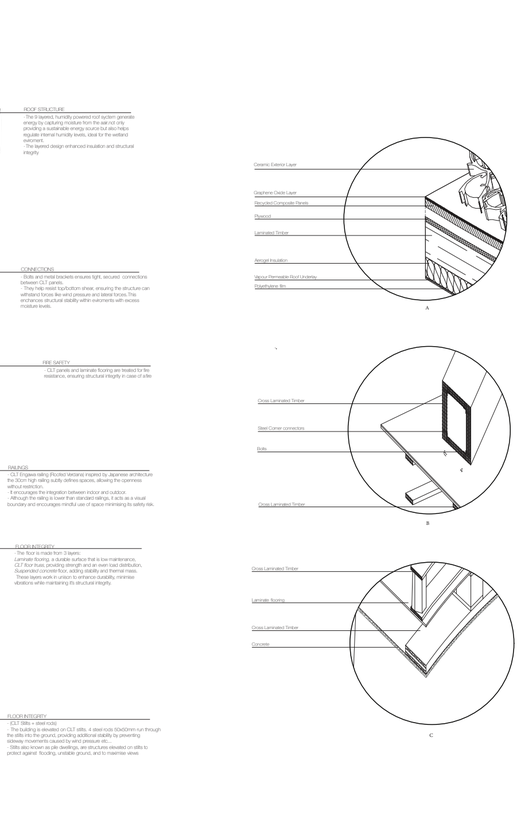






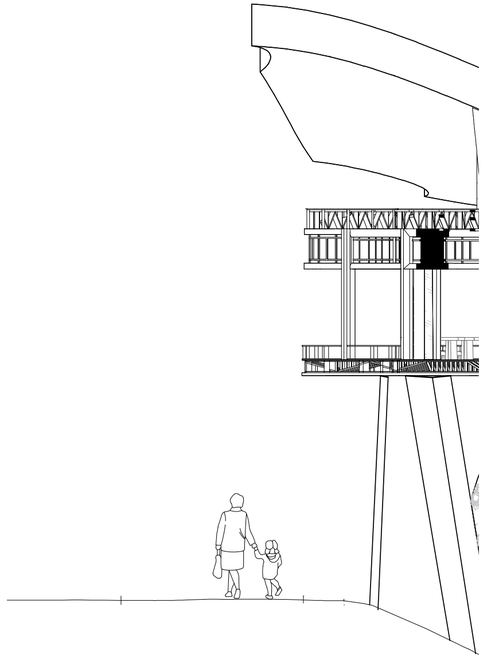
Due to its high surface area, oxygen-rich functional groups, mechanical properties, and strong moisture absorption capacity, GO demonstrates significant potential for sustainable energy generation. This material exploration investigates how atmospheric humidity can be transformed into an active architectural resource, combining environmental performance with innovative façade design.





Sunrise - *Walthamstow Wetlands*

Sunset - *Walthamstow Wetlands*

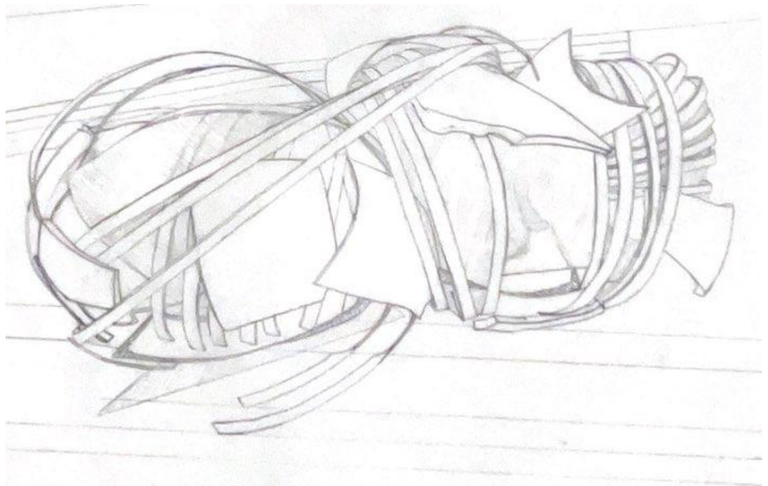


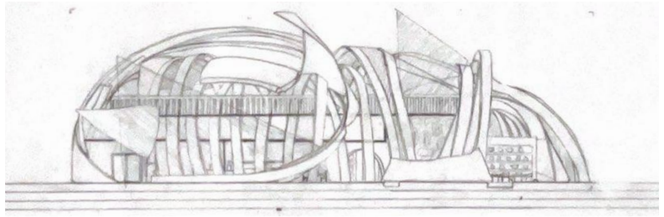
SOUTH-VIEW section

Potters Passage

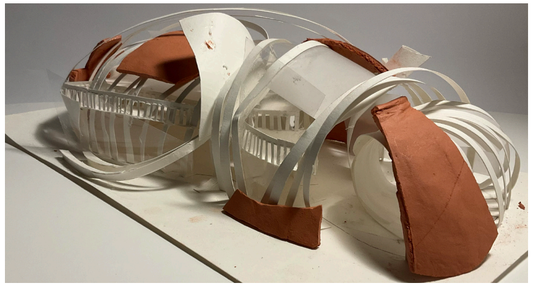
Pottery Studio

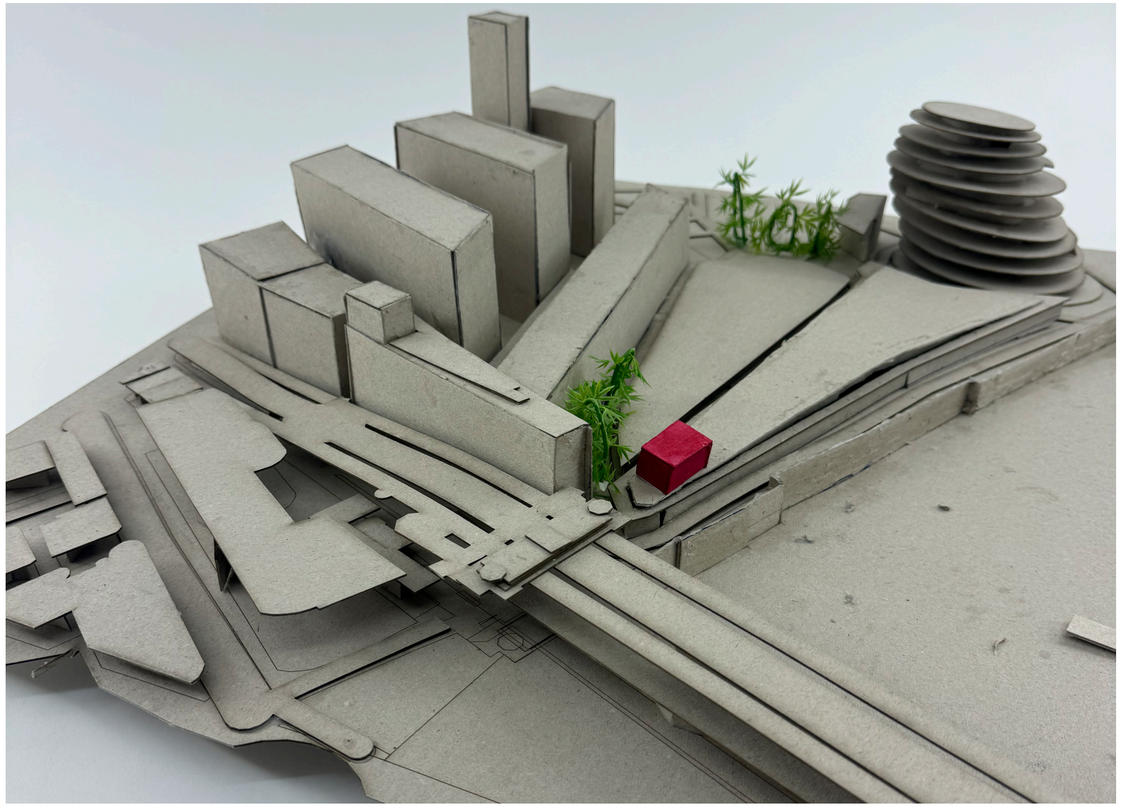
Rooted in the history of Potters Fields Park, this project transforms a former centre of ceramic production into a contemporary destination for making and cultural exchange. Inspired by the expressive forms of the Louis Vuitton Foundation, the architecture uses flowing geometry, natural light, and tactile materials to reflect the creative process of pottery while encouraging learning, exhibition, and community interaction.





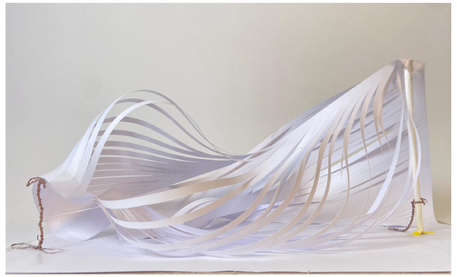
Front Elevation





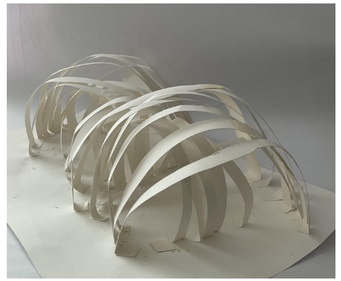
Site map - *Tower bridge*

Site sketches



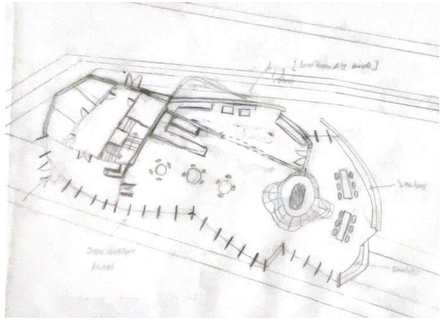
Experimental concept model

This page explores the manipulation of light through form. The experimental model uses folds to create curves and gaps that shape the way light passes through. Site sketches document areas where the architecture naturally manipulates light, informing the development of the model.

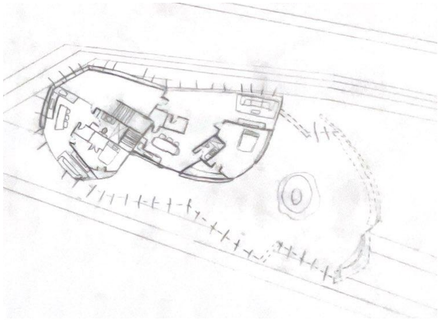


First iteration model

First iteration sketches

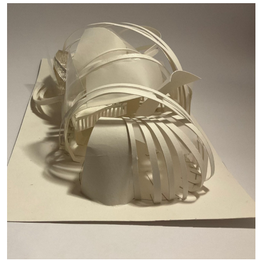
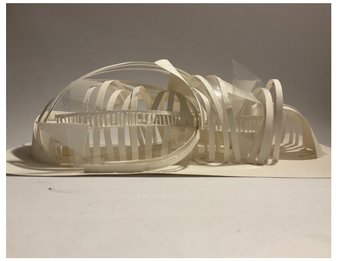


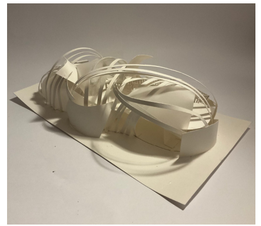
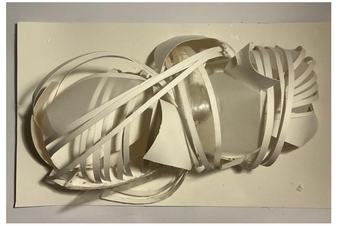
Ground floor



First floor

Precedent study - foundation louis vuitton

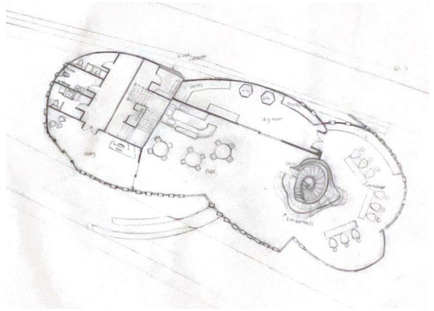



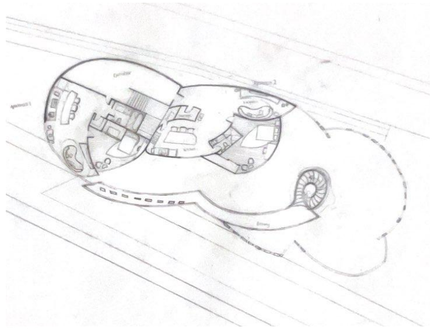
Developed iteration model

The precedent study of the Fondation Louis Vuitton influenced the development of the first design iteration. The roof was the primary focus, with its sail-like geometry inspiring the design.

developed iteration sketches



Ground floor



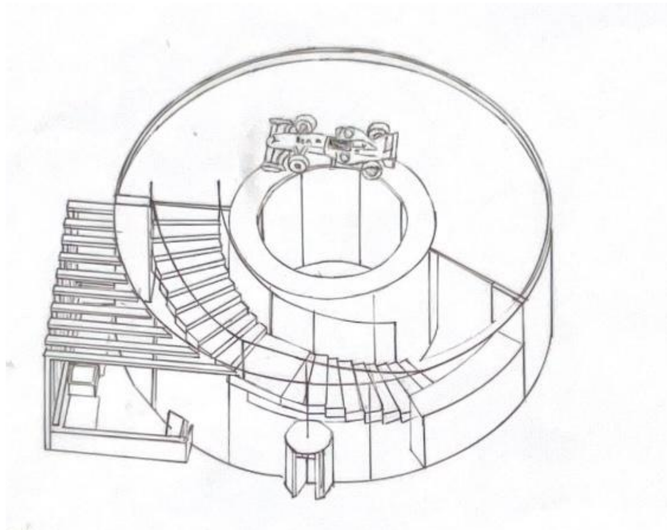
First floor

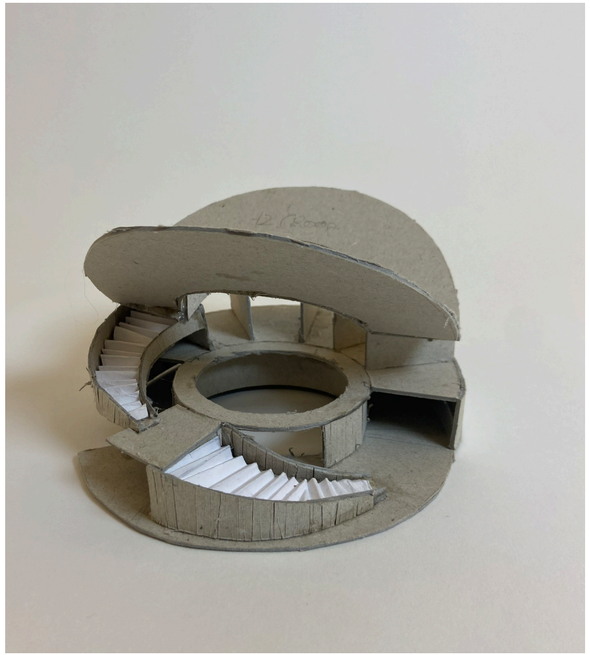
Potters passage - dry room

McLaren Velocity Centre

Formula 1 Experience Centre

Inspired by McLaren's Formula 1 legacy, this project translates the speed, precision, and innovation of motorsport into an interactive architectural destination. The building's flowing curves and aerodynamic forms reflect the profile of a Formula 1 tyre, creating a dynamic space that celebrates McLaren's heritage through exhibition, learning, and experiences.





initial concept model

Dry tyres

Red
The red soft slick tyre offers maximum grip and the fastest lap times on dry tracks but wears out quickly.

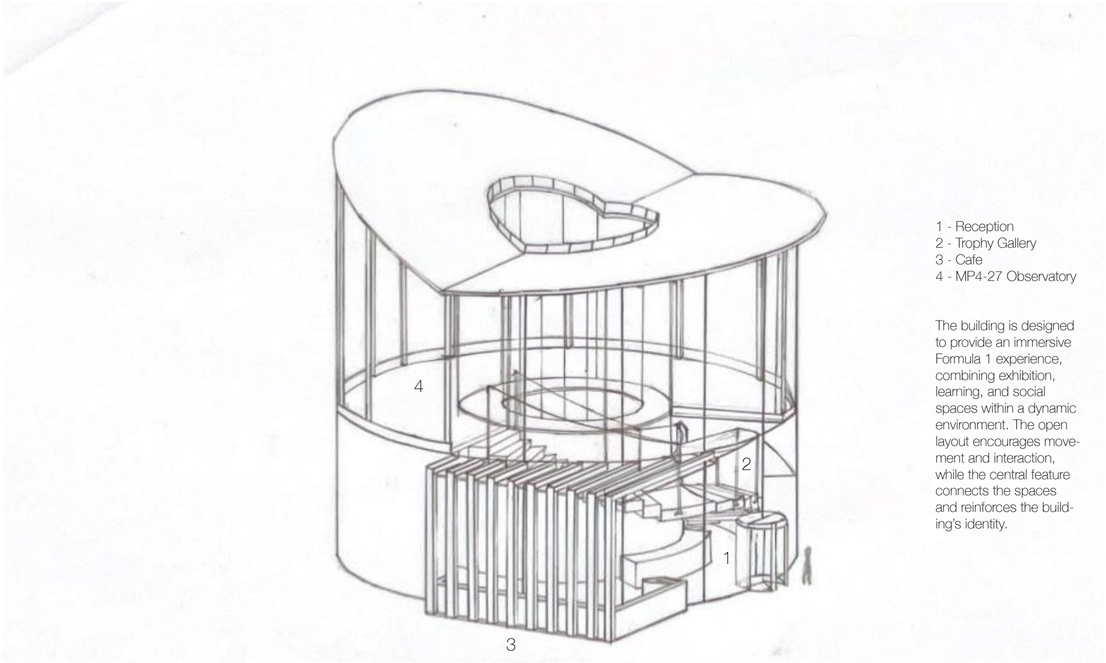
Yellow
The medium tyre balances grip and durability, combining strong performance with longer-lasting wear.

White
The hard tyre offers maximum durability and the longest lifespan but is the slowest tyre on the track.

Wet tyres

Green
The intermediate tyre is designed for damp or drying tracks, providing extra grip when full wet tyres aren't needed.

Blue
The full wet tyre is designed for heavy rain, providing maximum water dispersion and grip in very wet conditions.



1 - Reception
2 - Trophy Gallery
3 - Cafe
4 - MP4-27 Observatory

The building is designed to provide an immersive Formula 1 experience, combining exhibition, learning, and social spaces within a dynamic environment. The open layout encourages movement and interaction, while the central feature connects the spaces and reinforces the building's identity.

initial collage of cafe

McLaren Velocity centre - cafe

THANK YOU